

Gaussian beam migration

Angle migration

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Things we have learnt before (1)

Imaging condition:

$$\begin{aligned} I(\mathbf{x}) &= \int d\omega \frac{U(\mathbf{x}, \omega) D^*(\mathbf{x}, \omega)}{D(\mathbf{x}, \omega) D^*(\mathbf{x}, \omega) + \epsilon} \\ &\simeq \frac{\int d\omega U(\mathbf{x}, \omega) D^*(\mathbf{x}, \omega)}{\int d\omega D(\mathbf{x}, \omega) D^*(\mathbf{x}, \omega) + \epsilon} \end{aligned}$$

Incident (downgoing) waves:

$$D(\mathbf{x}) = G(\mathbf{x}, \mathbf{x}_s)$$

Scattered (upgoing) waves:

$$U(\mathbf{x}) = 2 \int \int dx_r dy_r P(\mathbf{x}_r) \partial_z^r G^*(\mathbf{x}, \mathbf{x}_r)$$

Things we have learnt before (2)

Change of variables to offset and midpoint:

$$\begin{aligned}\mathbf{h} &= \frac{1}{2}(\mathbf{x}_g - \mathbf{x}_s) = \text{offset} \\ \mathbf{x}_m &= \frac{1}{2}(\mathbf{x}_g + \mathbf{x}_s) = \text{midpoint}\end{aligned}$$

Common-offset (partial) image:

$$I(\mathbf{x}, \mathbf{h}) = K \int d\omega \int \int dx_m dy_m \frac{\partial G^*(\mathbf{x}_r, \mathbf{x}_m)}{\partial z_r} P(\mathbf{x}_m, \mathbf{h}) G^*(\mathbf{x}_m, \mathbf{x}_s)$$

Full stack (sum over offsets):

$$I(\mathbf{x}) = \int \int dh_x dh_y I(\mathbf{x}, \mathbf{h})$$

Next we will introduce Gaussian beam Green's functions in the equations above => Gaussian beam migration

Gaussian beam vs standard ray theory

Ray-theory Greens function:

$$g(\mathbf{x}, \mathbf{x}') = A(\mathbf{x}, \mathbf{x}') e^{i\omega T(\mathbf{x}, \mathbf{x}')}$$

Standard ray theory:

- T is real
- A is singular at the caustics

Gaussian beam:

- T is complex
- A is finite everywhere

There are many beam varieties:

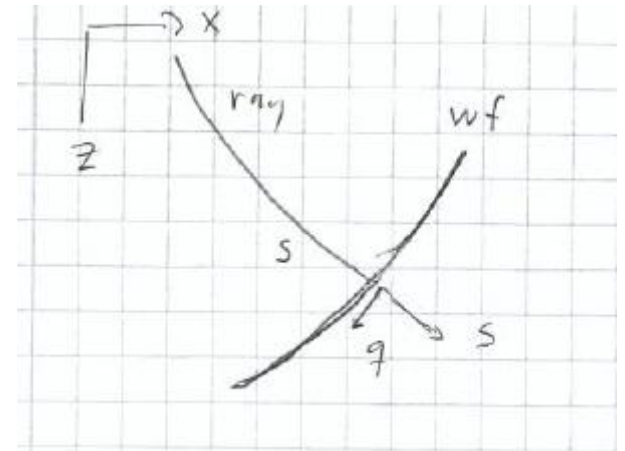
- Gaussian beam
- Controlled beam (CGG proprietary)
- Maslov asymptotic Greens function
- ...

Ray-centered coordinates (isotropic)

Assume τ is real part of traveltime

$$\mathbf{p} = \nabla \tau = |p|(0, 0, s)$$

because $\nabla \tau$ is normal to the wavefront



Complex traveltime in ray centered coordinates:

$$T(q_1, q_2, s) = \tau(s) + \frac{1}{2} q_I M_{IJ} q_J, \quad (I, J = 1, 2)$$

$$M_{IJ} = \frac{\partial^2 T}{\partial q_I \partial q_J}, \quad (I, J = 1, 2)$$

$$\Rightarrow \tau(s) = T(0, 0, s)$$

No linear term in T because $p \propto (0, 0, s)$

P and Q

Subdivision of matrix M

$$\mathbf{M} = \mathbf{P}^{-1}\mathbf{Q} = \mathbf{Q}_2$$

Dynamic ray tracing system (Hamilton):

$$\begin{aligned}\frac{\partial \mathbf{Q}}{\partial s} &\propto \mathbf{P} \\ \frac{\partial \mathbf{P}}{\partial s} &\propto \mathbf{Q}\end{aligned}$$

Gaussian beam: Choose complex $P(s_0)$:

$$\begin{aligned}\mathbf{P}(s_0) &= \frac{i}{v_0} \mathbf{I} \\ \mathbf{Q}(s_0) &= \frac{\omega w_l^2}{v_0} \mathbf{I}\end{aligned}$$

Then $\mathbf{Q}(s)$ is complex and non-singular
The beam width is w_l

$$g(x, x') = \sqrt{\frac{v(s)|\mathbf{Q}(s_0)|}{v(s_0)|\mathbf{Q}(s)|}} e^{i\omega[\tau(s) + \frac{1}{2}q_i M_{ij} q_j]}$$

Beam partial
Green's function

Standard RT: $|\det \mathbf{Q}_2| = 0$ at caustics

Gaussian beam Green's function (1)

Superposition of partial Green's functions:

$$G(\mathbf{x}, \mathbf{x}') = \int \int \Psi(\Omega) d\Omega g(\mathbf{x}, \mathbf{x}', \Omega)$$

$$\Psi(\Omega) \simeq \Psi_0 = \frac{i\omega}{2\pi v_0}$$

$$\Omega = (\theta, \phi) = \text{take-off direction}$$

Change of variables:

$$d\Omega = d\theta d\phi = \left| \frac{\partial(\theta, \phi)}{\partial(p_x, p_y)} \right| dp_x dp_y$$

Slowness and migration dip

Slowness:

$$p_x = \frac{1}{v} \sin \theta \cos \phi$$

$$p_y = \frac{1}{v} \sin \theta \sin \phi$$

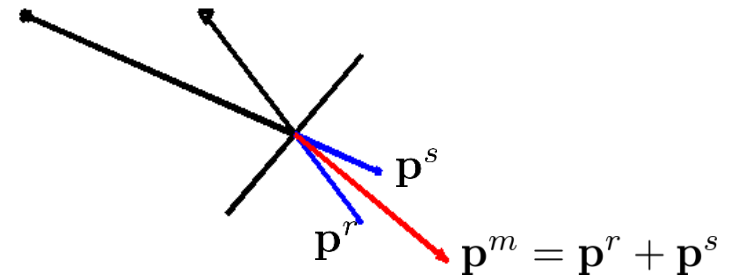
$$p_z = \frac{1}{v} \cos \theta$$

$$p_x^m = p_x^r + p_x^s$$

$$p_y^m = p_y^r + p_y^s$$

$$p_x^h = p_x^r - p_x^s$$

$$p_y^h = p_y^r - p_y^s$$



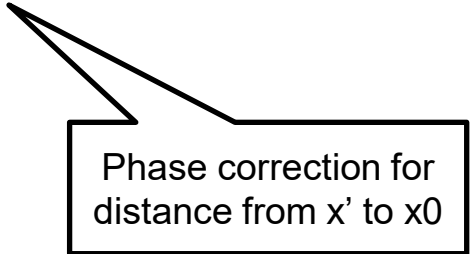
$p^m = p^r + p^s$ is the migration dip angle

Gaussian beam Green's function (2)

Superposition of partial Green's functions:

$$\begin{aligned} G(\mathbf{x}, \mathbf{x}') &= \frac{i\omega}{2\pi} \int \int \frac{dp'_x dp'_y}{p'_z} g(\mathbf{x}, \mathbf{x}', \mathbf{p}') \\ &= \frac{i\omega}{2\pi} \int \int \frac{dp'_x dp'_y}{p'_z} g(\mathbf{x}, \mathbf{x}_0, \mathbf{p}') e^{-i\omega p_i (\mathbf{x}'_i - \mathbf{x}_{0i})} \end{aligned}$$

$\mathbf{p}'$ = slowness at $\mathbf{x}'$
 $\mathbf{x}'$ = desired source point
 $\mathbf{x}_0$ = ray origin



Phase correction for
distance from $\mathbf{x}'$ to $\mathbf{x}_0$

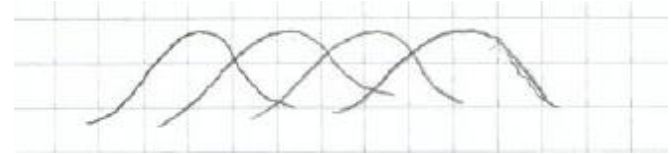
Gaussian packages (weights)

$$W_l(\mathbf{x}, \mathbf{x}_0) \propto e^{-\left|\frac{\omega}{\omega_l}\right| \frac{|\mathbf{x} - \mathbf{x}_0|^2}{2w_l^2}}$$

$$\sum W_l = 1$$

$$w_l = \text{beam width}$$

$$\omega_l = 2\pi v / w_l$$



Gaussian beam $\Rightarrow$ Maslov's asymptotic solution if $w_l \rightarrow \infty$

Gaussian beam migration

Common-offset (partial) image:

$$I(\mathbf{x}, \mathbf{h}) = K \int d\omega \int \int dx_m dy_m \frac{\partial G^*(\mathbf{x}_r, \mathbf{x})}{\partial z_r} P(\mathbf{x}_m, \mathbf{h}) G^*(\mathbf{x}, \mathbf{x}_s)$$

Substitute Gaussian beam GF in $I(\mathbf{x}, \mathbf{h})$:

$$I(\mathbf{x}, \mathbf{h}) = -C \sum_{\mathbf{x}_0} d\omega \int \int dp_x^m dp_y^m F(\mathbf{x}, \mathbf{x}_0, \mathbf{p}^m, \mathbf{h}) B(\mathbf{x}_0, \mathbf{p}^m, \mathbf{h})$$

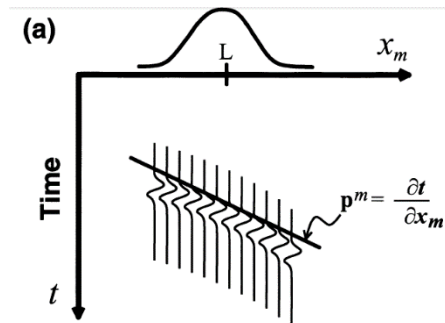
$$F(\mathbf{x}, \mathbf{x}_0, \mathbf{p}^m) = -\frac{i\omega}{2\pi} \int \int dp_x^h dp_y^h g^*(\mathbf{x}, \mathbf{x}_0 + \mathbf{h}, \mathbf{p}^r) g^*(\mathbf{x}, \mathbf{x}_0 - \mathbf{h}, \mathbf{p}^s)$$

$$B(\mathbf{x}_0, \mathbf{p}^m, \mathbf{h}) = \left| \frac{\omega}{\omega_l} \right|^3 \int \int \frac{dx' dy'}{4\pi} P(\mathbf{x}'_m, \mathbf{h}) e^{i\omega p_i^m (x'_i - x_{0i}) - \left| \frac{\omega}{\omega_l} \right| \frac{|\mathbf{x} - \mathbf{x}_0|^2}{2w_l^2}}$$

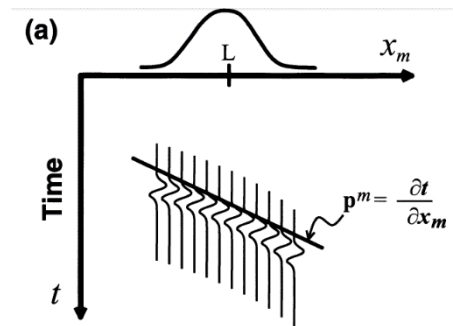
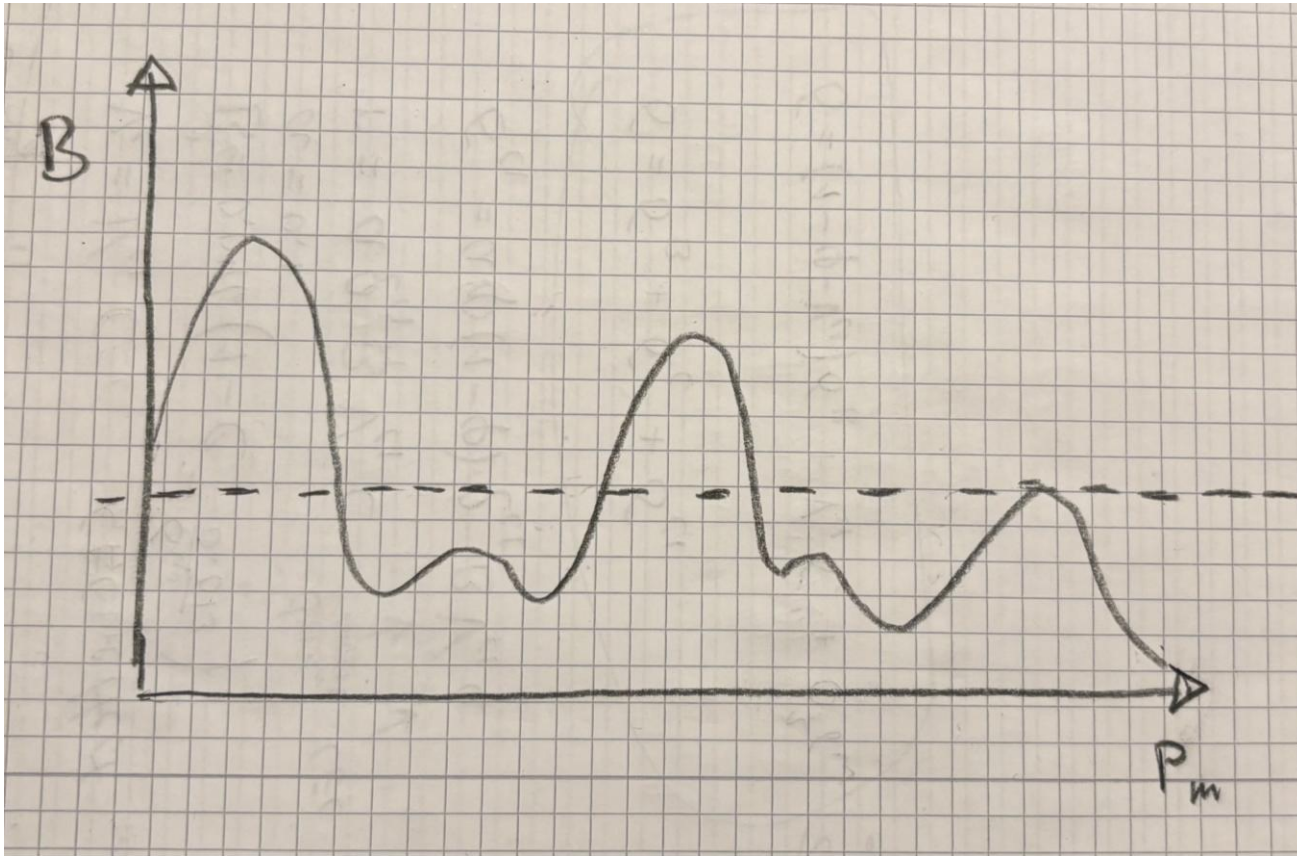
Beam propagation

Beam formation

- Beam formation (slant stack) can be done once and for all
- ... independent of velocity model
- Beam thresholding (subjective)



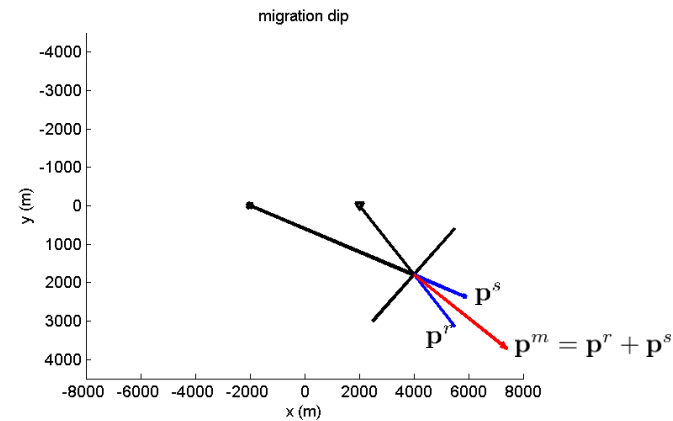
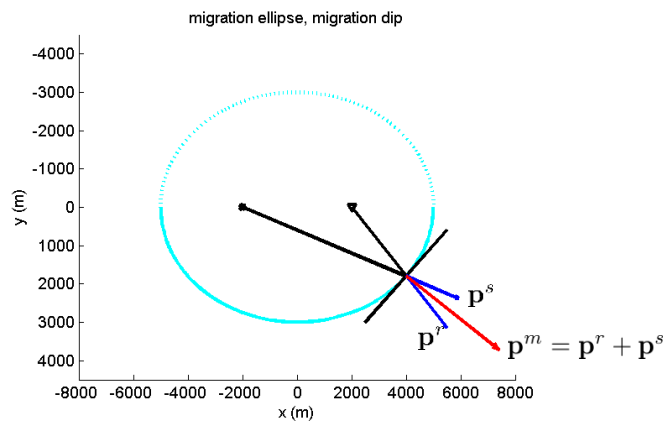
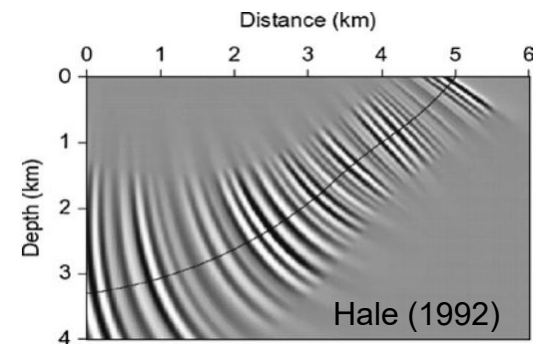
Beam thresholding



Slant stack and migration dip

Beam formation; slant stack:

$$B(x_0, p^m, h) = \left| \frac{\omega}{\omega_l} \right|^3 \iint \frac{dx' dy'}{4\pi} P(x'_m, h) e^{i\omega p_i^m (x'_i - x_{0i}) - \left| \frac{\omega}{\omega_l} \right| \frac{|x - x_0|^2}{2w_l^2}}$$



From Gray et al., TLE (2009)

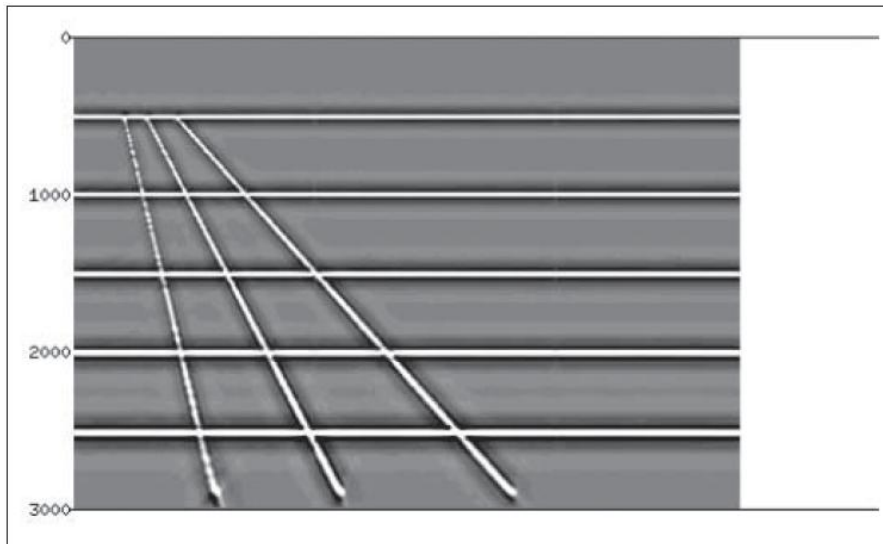


Figure 1. Prestack Kirchhoff depth-migrated image of a fan model data set, combining flat reflectors and dipping reflectors with dips of 45° , 60° , and 75° .

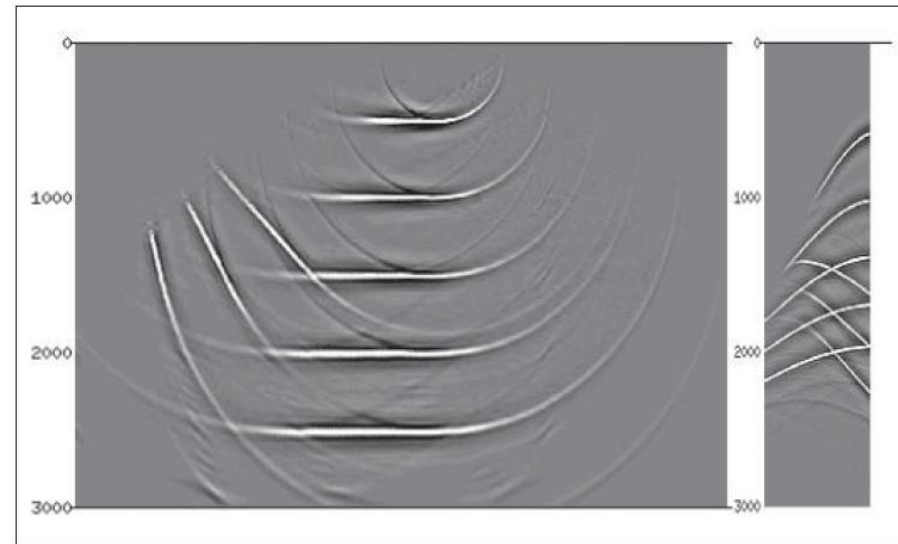


Figure 2. (left) Kirchhoff depth-migrated image of a single shot selected from the middle of the survey. (right) The unmigrated shot record.

From Gray et al., TLE (2009)

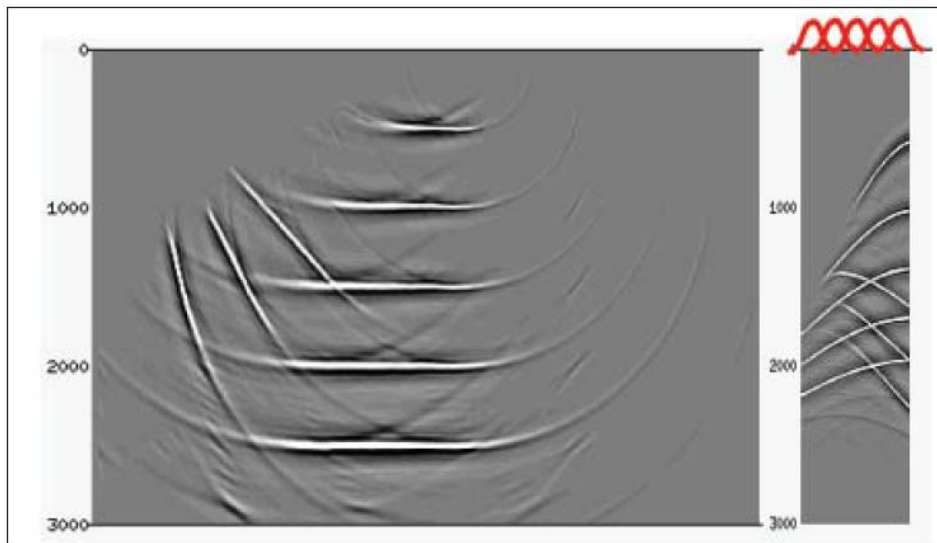


Figure 4. Gaussian-beam migration of the same shot as in Figure 2. The image is slightly cleaner than the corresponding Kirchhoff image, and has fewer nongeologic swings. The red curves over the unmigrated shot record indicate data decomposition into overlapping regions, handled separately by the migration.

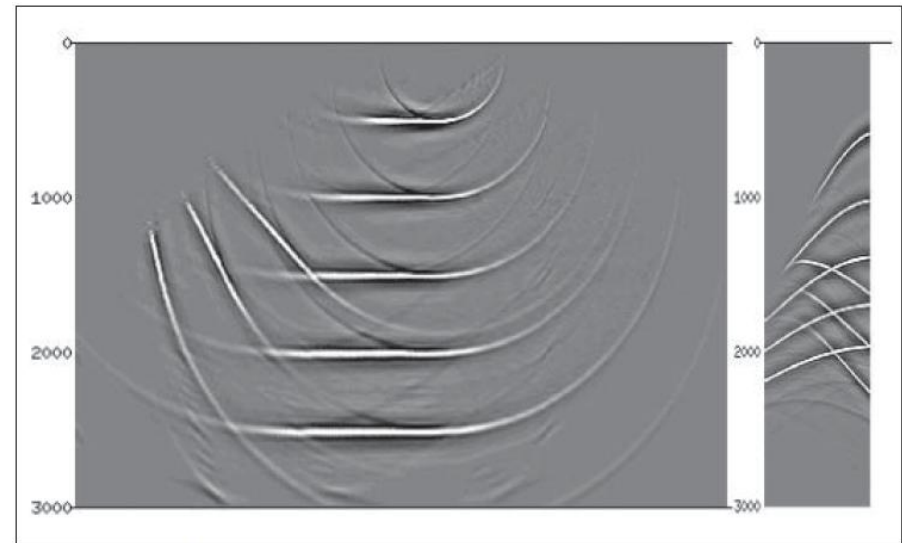


Figure 2. (left) Kirchhoff depth-migrated image of a single shot selected from the middle of the survey. (right) The unmigrated shot record.

From Gray et al., TLE (2009)

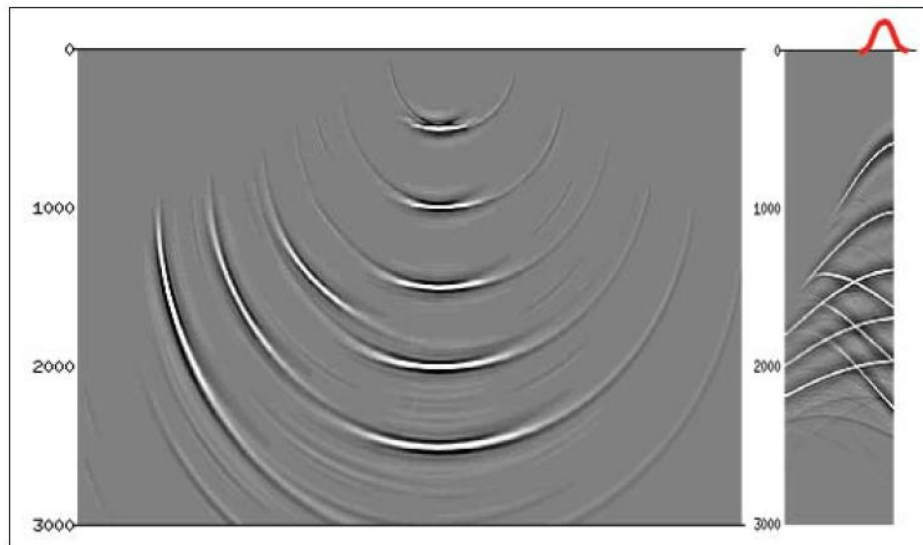
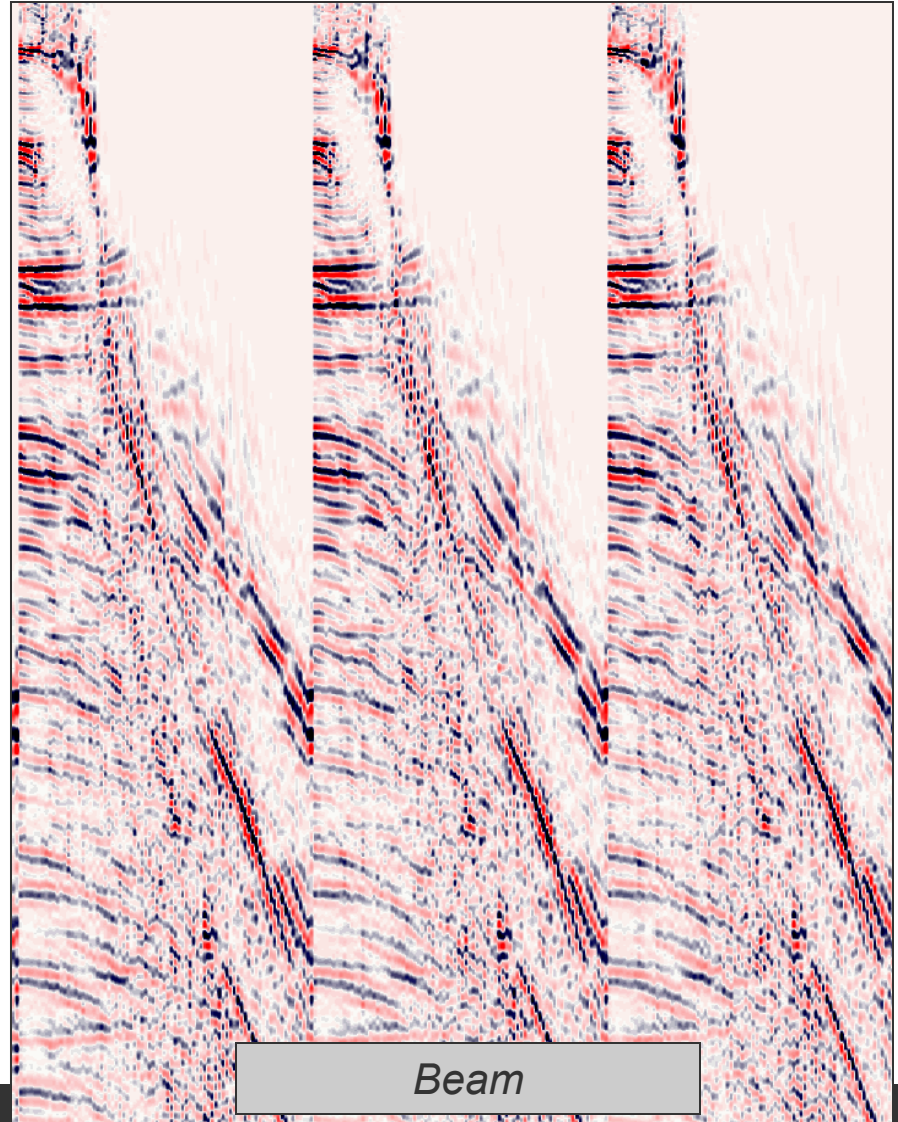
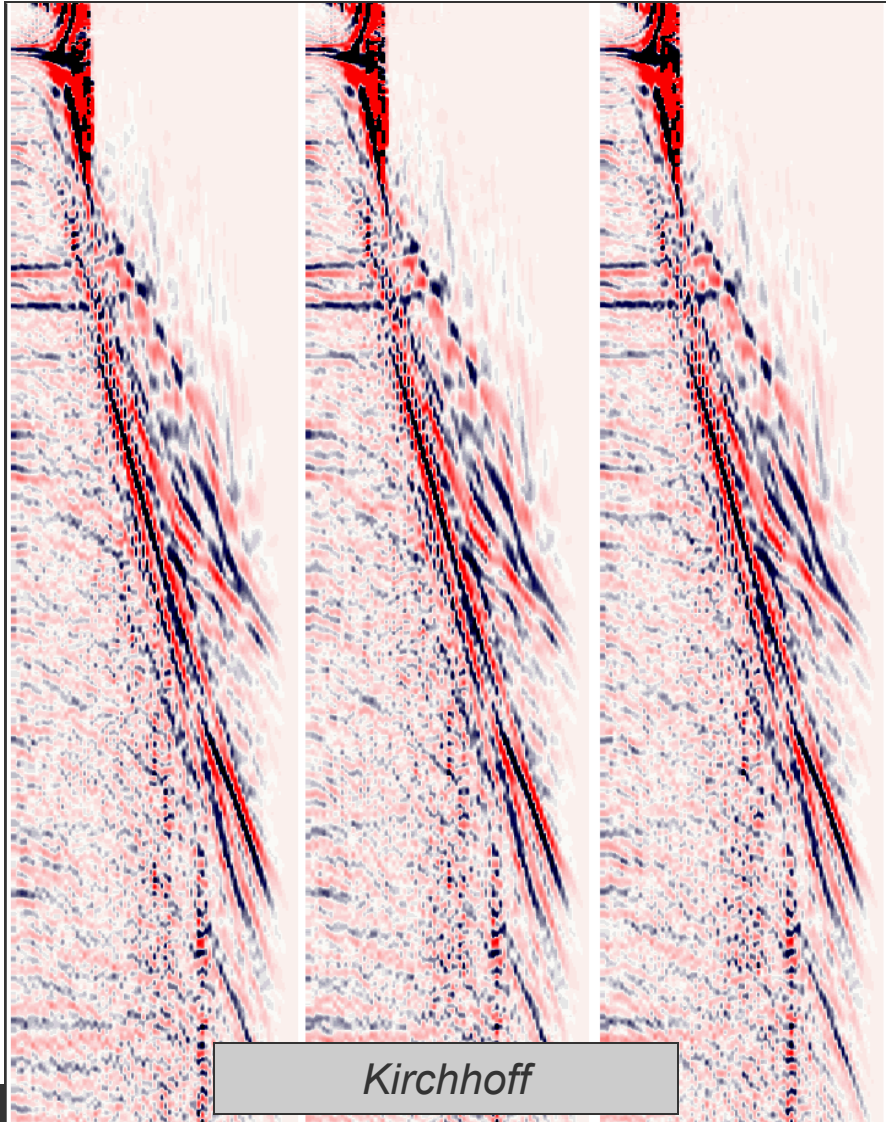


Figure 5. Gaussian-beam migration of the traces in the region nearest the shotpoint.

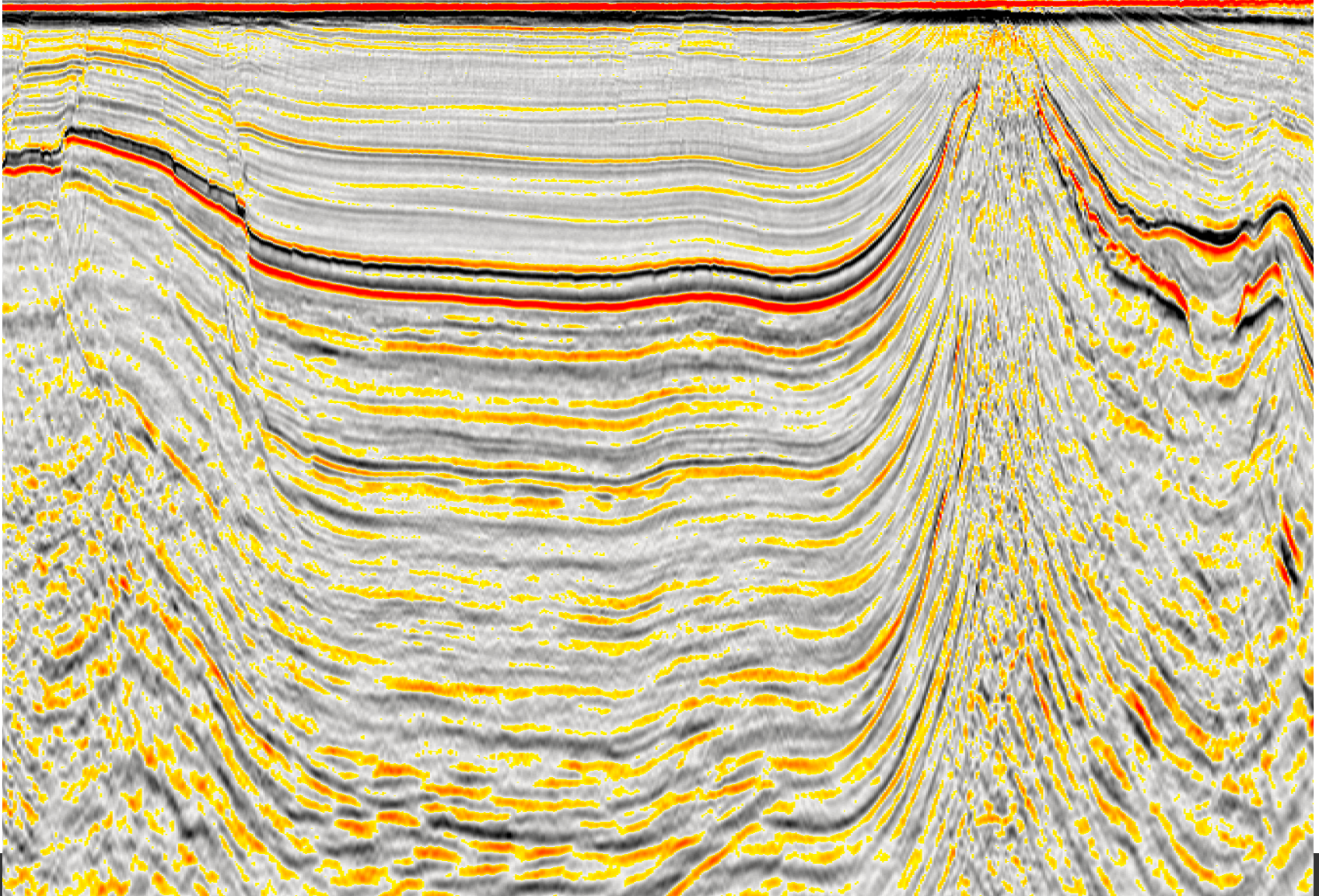


Figure 3. Kirchhoff depth-migrated image of a single near-offset trace from the same shot migrated in Figure 2. This is the basic molecule of Kirchhoff migration.

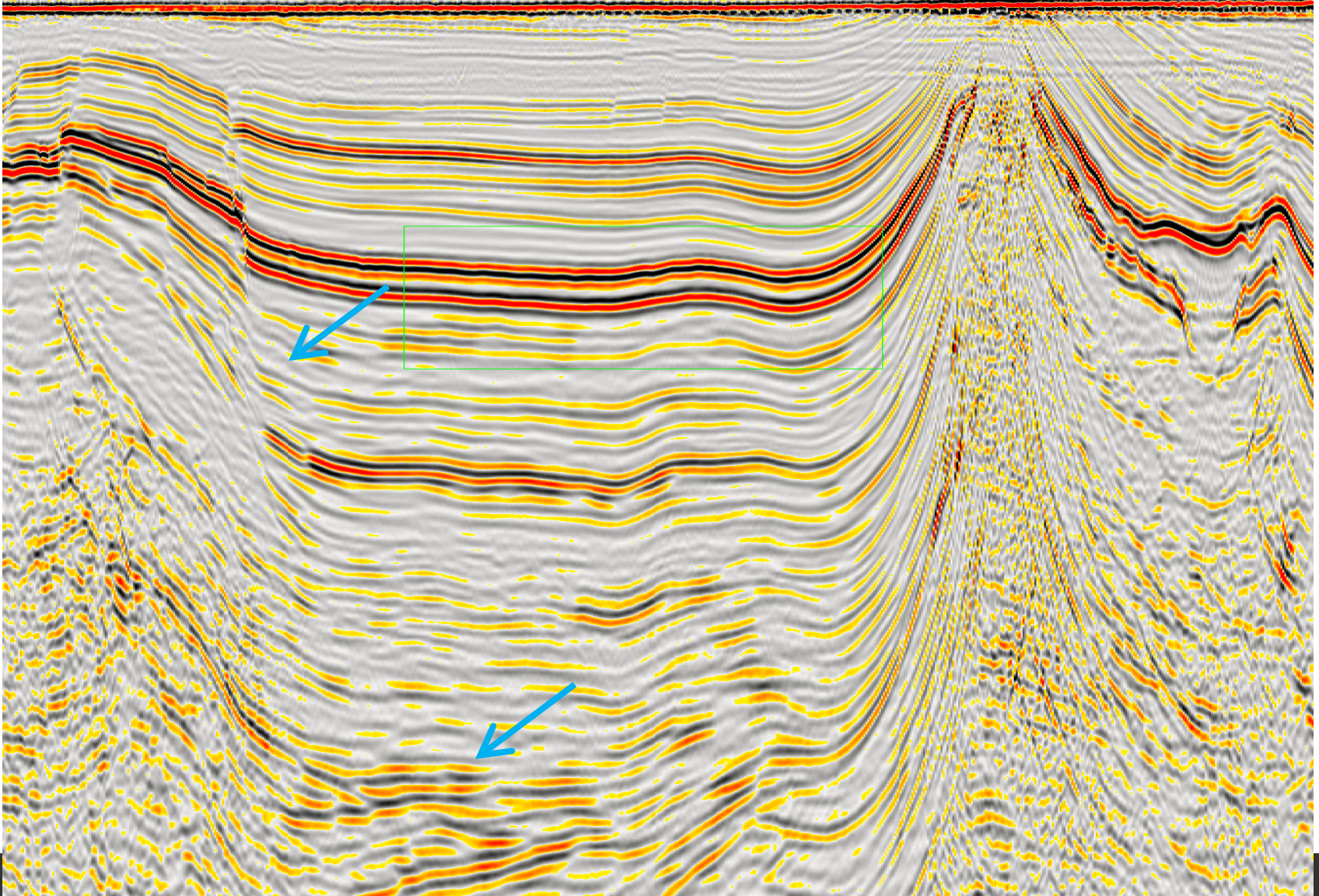
ST0811 – CIP gathers after Depth Migration



ST0811 PSDM - TTI Kirchhoff migration



ST0811 PSDM - TTI Beam migration



Angle migration

Basic idea of angle migration

$$\int \int \int \int dh_x dh_y dx_m dy_m \rightarrow \int \int \int \int \left| \frac{\partial(h_x, h_y, x_m, y_m)}{\partial(\theta, \phi, \nu_1, \nu_2)} \right| d\theta d\phi d\nu_1 d\nu_2$$

Change of variables from surface coordinates to scattering angle and dip

Common-angle partial image:

$$I(\mathbf{x}, \theta, \phi) = \int \int J d\nu_1 d\nu_2 W(\cdot) \partial_t P(\mathbf{h}(\nu_1, \nu_2), \mathbf{x}_m(\nu_1, \nu_2), \tau)$$

Beylkin determinant (non-trivial):

$$J = \left| \frac{\partial(h_x, h_y, x_m, y_m)}{\partial(\theta, \phi, \nu_1, \nu_2)} \right|$$

Mapping from scattering angle and dip to surface coordinates.

Angle migration – integration over migration dip

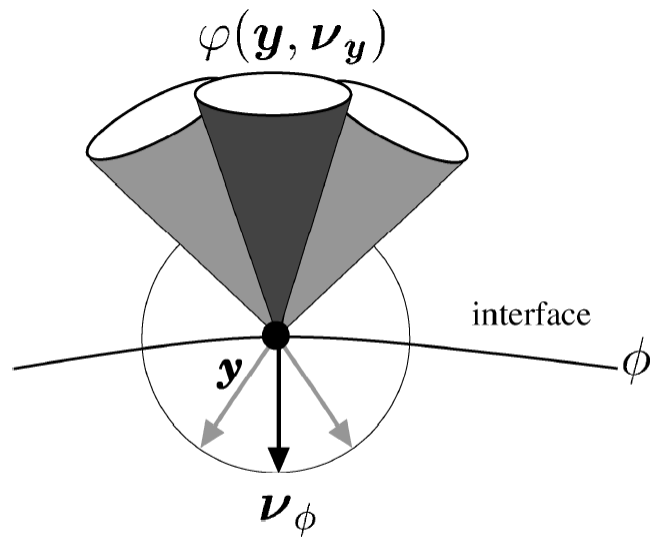
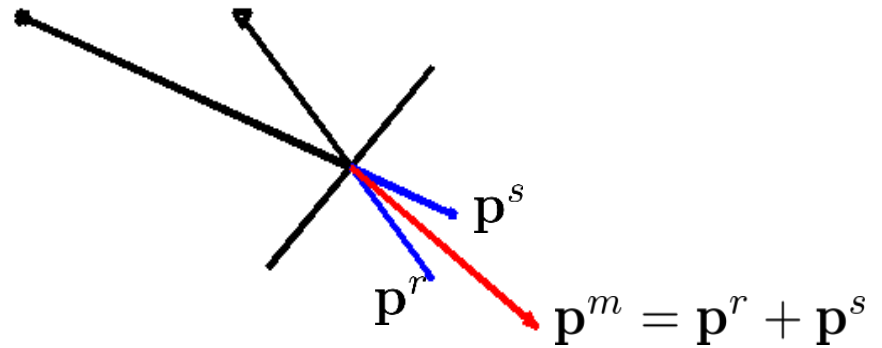


FIG. 2. Focusing in dip: partitioning of the integration over migration dip.

(Brandsberg-Dahl et al., Geophysics, 2003)



$p^m = p^s + p^r$ is the migration dip angle

Sverre's synthetic gas-cloud example

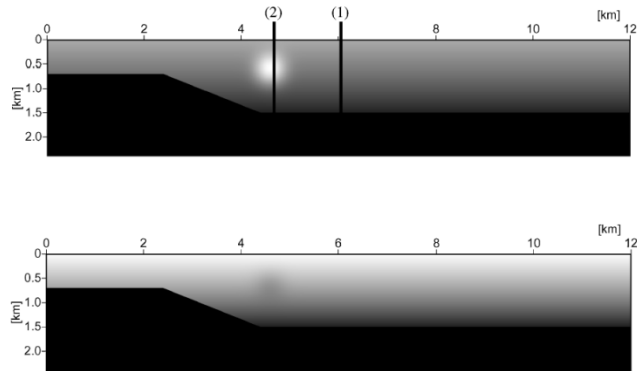


FIG. 4. P-wave and S-wave velocity profiles for the gas-cloud model. Also shown are the two locations where CIGs were produced.

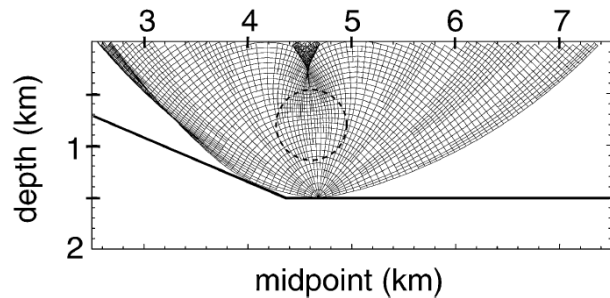
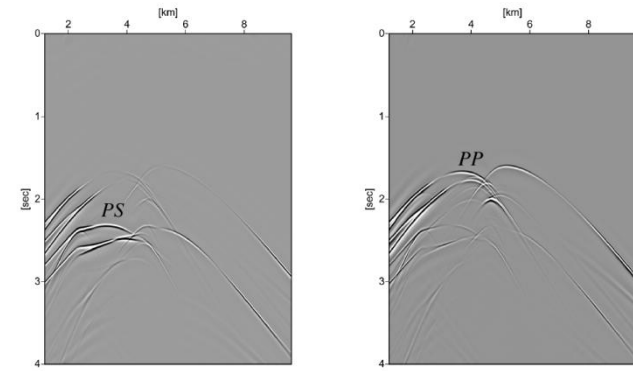


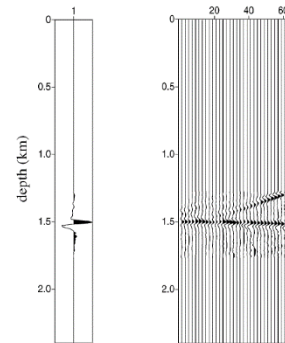
FIG. 5. A ray fan from the gas-cloud P-wave velocity model at position 2.



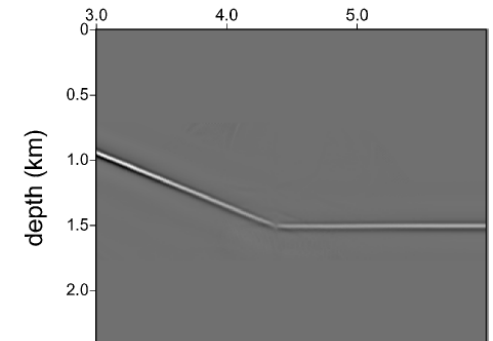
(a) x-component

(b) z-component

FIG. 6. A common-shot gather at location 2 from the finite-difference modeling in the gas-cloud model. The x-axis indicates the true receiver location in the model.



(d) Angle gather and stack for the sum of the three cones

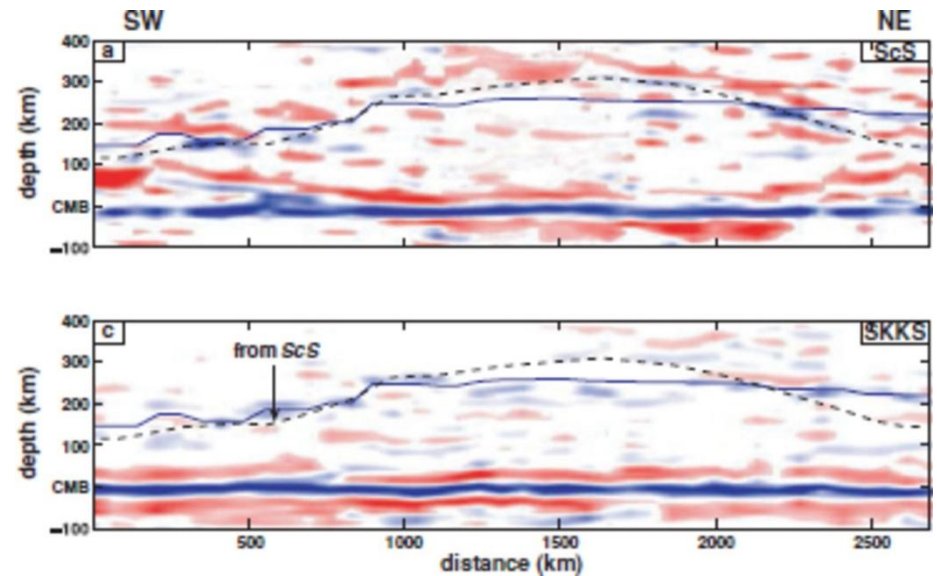
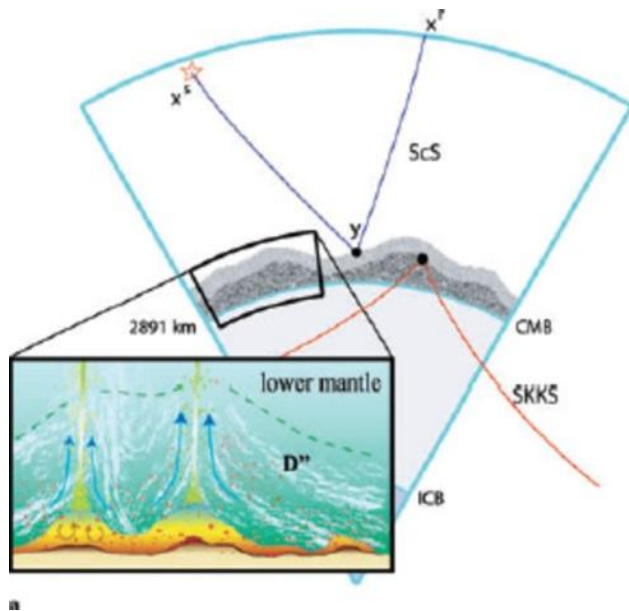


(d) image focused in reflector dip

(Brandsberg-Dahl et al., Geophysics, 2003)

Angle migration example

Angle migration (GRT)



(Ping Wang et al. Geophys. J. Int. 2008)